

Unit 2: First Order Linear Differential Equations

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 2 at a Glance: Core Sub-Topics

This unit assembles the core solving toolkit of first order theory. Each sub-topic below builds on the last, moving from raw integration techniques toward genuine physical models.

- **Separable Equations** — splitting a rate law into a pure x side and a pure y side, then integrating each independently.
- **First-Order Linear Equations and Integrating Factors** — the standard form $y' + P(x)y = Q(x)$ and the factor that collapses the left side into a single derivative.
- **Initial Value Problems** — using a known data point to pin the arbitrary constant and select one curve from the solution family.
- **Homogeneous Equations and Substitution** — the substitution $y = vx$ that reduces a degree-matched equation to a separable one.
- **Applications: Growth, Decay, and Newton's Law of Cooling** — the exponential models that these methods unlock.

Part I — Condensed Cheat Sheet

2.1 Separable Equations

A first order equation is *separable* when its right side factors into a function of x alone times a function of y alone — that single structural test is the whole game. Once it factors, you divide and multiply until every y rides with dy and every x rides with dx , then integrate both sides.

Standard Form and Method: Separable

A separable equation and its solution scheme:

$$\frac{dy}{dx} = g(x) h(y) \implies \int \frac{1}{h(y)} dy = \int g(x) dx + C.$$

One constant C suffices, since the two constants of integration merge into a single arbitrary constant.

Pitfall. Dividing by $h(y)$ silently assumes $h(y) \neq 0$. Any constant value of y that makes $h(y) = 0$ is an equilibrium solution that this step can quietly discard — always check it separately.

Worked Example 2.1: Solve $\frac{dy}{dx} = 6x^2y$

Step 1. Separate the variables:

$$\frac{1}{y} dy = 6x^2 dx.$$

Step 2. Integrate both sides:

$$\ln |y| = 2x^3 + C.$$

Step 3. Exponentiate to isolate y :

$$y = e^{2x^3+C} = C e^{2x^3}.$$

The dropped solution $y = 0$ is recovered as the case $C = 0$.

2.2 First-Order Linear Equations and Integrating Factors

A first order equation is *linear* when y and y' each appear only to the first power and never multiply one another. Before reading off anything, you must normalize so the derivative carries coefficient one — only then is $P(x)$ truly the coefficient of y .

Standard Form and Integrating Factor

Standard form, integrating factor, and solution:

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \mu(x) = e^{\int P(x) dx},$$

$$\frac{d}{dx} [\mu(x)y] = \mu(x)Q(x) \implies y = \frac{1}{\mu(x)} \left(\int \mu(x)Q(x) dx + C \right).$$

Algorithm: Solving a First-Order Linear Equation

1. Write the equation in standard form $y' + P(x)y = Q(x)$, dividing through so the derivative carries coefficient one.
2. Compute the integrating factor $\mu(x) = e^{\int P(x) dx}$.
3. Multiply through by $\mu(x)$; the left side collapses to the single derivative $\frac{d}{dx}[\mu(x)y]$.
4. Integrate both sides, then divide by $\mu(x)$ to isolate y .

The magic of $\mu(x)$ is that multiplying through turns the entire left side into the exact product-rule derivative $\frac{d}{dx}[\mu y]$ — that is the single fact the whole method rests on.

Worked Example 2.2: Solve $\frac{dy}{dx} + y = e^x$

Step 1. Read off $P(x) = 1$, so the integrating factor is

$$\mu = e^{\int 1 dx} = e^x.$$

Step 2. Multiply through; the left side collapses to a derivative:

$$\frac{d}{dx}[e^x y] = e^x \cdot e^x = e^{2x}.$$

Step 3. Integrate both sides:

$$e^x y = \frac{1}{2}e^{2x} + C.$$

Step 4. Divide by e^x :

$$y = \frac{1}{2}e^x + Ce^{-x}.$$

2.3 Initial Value Problems

The general solution of a first order equation is a whole family of curves indexed by one arbitrary constant. An *initial condition* $y(x_0) = y_0$ selects exactly one member — it fixes the constant to the value that forces the curve through the given point.

General versus Particular Solution

A first order equation needs exactly **one** initial condition. The general solution carries the free constant; the particular solution has it determined:

$$y = F(x) + C \xrightarrow{y(x_0)=y_0} C = y_0 - F(x_0).$$

Worked Example 2.3: Solve $\frac{dy}{dx} = 3x^2$, $y(1) = 5$

Step 1. Integrate for the general solution:

$$y = x^3 + C.$$

Step 2. Apply $y(1) = 5$:

$$5 = (1)^3 + C \implies C = 4.$$

Step 3. State the particular solution:

$$y = x^3 + 4.$$

2.4 Homogeneous First-Order Equations and Substitution

A function $f(x, y)$ is *homogeneous of degree n* when scaling both inputs pulls out a clean power of the scale factor: $f(tx, ty) = t^n f(x, y)$. An equation $M dx + N dy = 0$ is homogeneous when M and N share the same degree — and that shared degree is precisely what lets the ratio substitution work.

The Substitution $y = vx$

For a homogeneous equation, set

$$y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx},$$

which always reduces the equation to a *separable* one in v and x . After solving for v , back-substitute $v = \frac{y}{x}$.

Worked Example 2.4: Solve $\frac{dy}{dx} = \frac{x+2y}{x}$

Step 1. Rewrite the right side and substitute $y = vx$:

$$\frac{dy}{dx} = 1 + 2\frac{y}{x} = 1 + 2v, \quad v + x \frac{dv}{dx} = 1 + 2v.$$

Step 2. Cancel v and separate:

$$x \frac{dv}{dx} = 1 + v \implies \frac{1}{1+v} dv = \frac{1}{x} dx.$$

Step 3. Integrate and exponentiate:

$$\ln |1+v| = \ln |x| + C \implies 1+v = Cx.$$

Step 4. Back-substitute $v = \frac{y}{x}$ and clear:

$$1 + \frac{y}{x} = Cx \implies y = Cx^2 - x.$$

2.5 Applications: Growth, Decay, and Newton's Law of Cooling

When a rate of change is proportional to the current amount, the solution is exponential — this single idea drives population models, radioactive decay, and cooling alike.

Exponential Models

Growth and decay:

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt} \quad (k > 0 \text{ grows, } k < 0 \text{ decays}),$$

with doubling time $t = \frac{\ln 2}{k}$. Newton's law of cooling:

$$\frac{dT}{dt} = -k(T - T_s) \implies T(t) = T_s + (T_0 - T_s)e^{-kt},$$

so $T \rightarrow T_s$ as $t \rightarrow \infty$.

Worked Example 2.5: A culture doubles in 3 hours

Step 1. The model $P = P_0 e^{kt}$ doubles when $e^{3k} = 2$:

$$3k = \ln 2 \implies k = \frac{\ln 2}{3}.$$

Step 2. The growth law is therefore

$$P(t) = P_0 e^{(\ln 2/3)t} = P_0 2^{t/3}.$$

Every 3 hours the population multiplies by two, as required.

Part II — Review Guide: Practice Problems

Work the following ten problems in order; they are graded from direct separation through linear factors, initial value problems, the homogeneous substitution, and a full physical model. Solutions follow in Part III.

1. Solve $\frac{dy}{dx} = \frac{x}{y}$ by separation of variables.
 2. Solve $\frac{dy}{dx} = e^{x-y}$.
 3. Solve $\frac{dy}{dx} = y \cos x$, and identify any solution that is lost when you divide by y .
 4. Solve the linear equation $\frac{dy}{dx} + 3y = 6$ using an integrating factor.
 5. Solve $x \frac{dy}{dx} + y = x^2$ by first writing it in standard linear form.
 6. Solve $\frac{dy}{dx} + \frac{2}{x}y = x$.
 7. Solve the initial value problem $\frac{dy}{dx} + 2y = 4$ with $y(0) = 1$.
 8. Solve the homogeneous equation $\frac{dy}{dx} = \frac{x+y}{x}$ using the substitution $y = vx$.
 9. Solve the homogeneous equation $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ using $y = vx$.
 10. **(Application)** An object at 100° is placed in a room held at 20° . After 10 minutes its temperature is 60° . Find the temperature function $T(t)$ and determine how long it takes the object to reach 30° .
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Part III — Complete Solutions

Solution 1. Separate, then integrate.

$$\begin{aligned}y \, dy &= x \, dx \\ \frac{y^2}{2} &= \frac{x^2}{2} + C_1 \\ y^2 &= x^2 + C \quad (\text{general solution}).\end{aligned}$$

Solution 2. Split the exponential as $e^{x-y} = e^x e^{-y}$, then separate.

$$\begin{aligned}e^y \, dy &= e^x \, dx \\ e^y &= e^x + C \\ y &= \ln(e^x + C).\end{aligned}$$

Solution 3. Separate (assuming $y \neq 0$), integrate, and exponentiate.

$$\begin{aligned}\frac{1}{y} \, dy &= \cos x \, dx \\ \ln |y| &= \sin x + C \\ y &= C e^{\sin x}.\end{aligned}$$

Dividing by y assumed $y \neq 0$, so it dropped the equilibrium solution $y = 0$ — which is recovered as the special case $C = 0$.

Solution 4. Here $P = 3$, so the integrating factor is $\mu = e^{\int 3 \, dx} = e^{3x}$. Multiply through:

$$\begin{aligned}\frac{d}{dx}[e^{3x}y] &= 6e^{3x} \\ e^{3x}y &= \int 6e^{3x} \, dx = 2e^{3x} + C \\ y &= 2 + Ce^{-3x}.\end{aligned}$$

Solution 5. Divide by x to reach standard form:

$$\frac{dy}{dx} + \frac{1}{x}y = x, \quad \mu = e^{\int (1/x) \, dx} = e^{\ln|x|} = x.$$

Multiply through and integrate:

$$\begin{aligned}\frac{d}{dx}[xy] &= x \cdot x = x^2 \\ xy &= \frac{x^3}{3} + C \\ y &= \frac{x^2}{3} + \frac{C}{x}.\end{aligned}$$

Solution 6. With $P = \frac{2}{x}$, integrate the coefficient and exponentiate:

$$\mu = e^{\int (2/x) dx} = e^{2 \ln |x|} = x^2.$$

Multiply through and integrate:

$$\frac{d}{dx}[x^2 y] = x^2 \cdot x = x^3$$

$$x^2 y = \frac{x^4}{4} + C$$

$$y = \frac{x^2}{4} + \frac{C}{x^2}.$$

Solution 7. The integrating factor is $\mu = e^{\int 2 dx} = e^{2x}$. Multiply through and integrate:

$$\frac{d}{dx}[e^{2x} y] = 4 e^{2x}$$

$$e^{2x} y = 2 e^{2x} + C$$

$$y = 2 + C e^{-2x}.$$

Apply the initial condition $y(0) = 1$:

$$1 = 2 + C \implies C = -1$$

$$y = 2 - e^{-2x}.$$

Solution 8. Rewrite as $\frac{dy}{dx} = 1 + \frac{y}{x}$ and substitute $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$v + x \frac{dv}{dx} = 1 + v$$

$$x \frac{dv}{dx} = 1 \implies dv = \frac{1}{x} dx$$

$$v = \ln |x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$y = x \ln |x| + Cx.$$

Solution 9. Substitute $y = vx$. The right side becomes

$$\frac{x^2 + v^2 x^2}{x \cdot vx} = \frac{1 + v^2}{v}.$$

With $\frac{dy}{dx} = v + x \frac{dv}{dx}$,

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{v} = \frac{1}{v} + v$$

$$x \frac{dv}{dx} = \frac{1}{v} \implies v dv = \frac{1}{x} dx$$

$$\frac{v^2}{2} = \ln |x| + C_1 \implies v^2 = 2 \ln |x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$\frac{y^2}{x^2} = 2 \ln |x| + C \implies y^2 = x^2 (2 \ln |x| + C).$$

Solution 10. Newton's law of cooling with surrounding temperature $T_s = 20$ and initial temperature $T_0 = 100$ gives

$$T(t) = 20 + (100 - 20) e^{-kt} = 20 + 80 e^{-kt}.$$

Use the reading $T(10) = 60$ to find k :

$$60 = 20 + 80 e^{-10k} \implies 80 e^{-10k} = 40$$

$$e^{-10k} = \frac{1}{2} \implies -10k = -\ln 2 \implies k = \frac{\ln 2}{10}.$$

Hence the temperature function is

$$T(t) = 20 + 80 e^{-(\ln 2/10)t} = 20 + 80 \left(\frac{1}{2}\right)^{t/10}.$$

To reach 30° , solve $T(t) = 30$:

$$30 = 20 + 80 e^{-kt} \implies 80 e^{-kt} = 10 \implies e^{-kt} = \frac{1}{8}$$

$$-kt = -\ln 8 = -3 \ln 2 \implies t = \frac{3 \ln 2}{k} = \frac{3 \ln 2}{\ln 2/10} = 30 \text{ minutes}.$$

Unit 2 Key Takeaway

Every method in this unit reduces to a single move — rewrite the equation into a form you can integrate. **Separation of variables** splits a factored rate law; the **integrating factor** $\mu = e^{\int P dx}$ collapses a linear left side into one derivative; and the **substitution** $y = vx$ turns a homogeneous equation separable. Read the structure first, and the technique always follows.